SKYE8 COMPANY LTD

Data Science and ML / AI Internship Track

FINAL PROJECT

The Live Voice Conversation System

Issued to	MOFOR ROCHELLE
Roadmap reference	SKY8-DS-RM-006 (Core ML)
Project reference	SKY8-DS-FP-006
Duration	Seven days - issued Friday 25 September 2026, closes 23:59 Friday 2 October 2026
Repository	skye8-ds-fp-mofor-rochelle
Submission	Skye8 Platform
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1. The Product

Open Gemini Live or a similar assistant, press the button, and you talk to it the way you talk to a person. You speak, it answers in a voice, captions appear as it goes, and if you cut in while it is still talking it stops and listens. It feels like a call, not like sending a message and waiting.

You are building that. An English voice assistant a user can hold a real conversation with: press to start, speak naturally, hear a spoken reply with captions on screen, and interrupt at any point.

What actually makes it feel live. Not the speech recognition and not the language model. Both of those you may call. What separates a system that feels like a conversation from one that feels broken is knowing when the person has finished speaking. Cut in too early and you talk over them mid-sentence. Wait too long and there is an awkward silence and the user repeats themselves. That decision has to be made in a fraction of a second, hundreds of times per conversation.

Why this needs machine learning. A silence threshold cannot solve it. People pause in the middle of sentences to think, and they finish sentences without any pause at all. Telling a thinking pause apart from a finished turn requires a model that uses how the voice is falling, how long the pause has lasted, and what has been said so far. That model is what you are building. Everything else in this project is the wrapper around it.

Who buys it. Businesses that answer the same questions by phone all day - clinics, schools, delivery services, customer support lines. Also anyone building a voice product for users who would rather speak than type, which in Cameroon is most users.

2. Data

Source	What it gives you
AMI Meeting Corpus	Free from the University of Edinburgh. 100 hours of recorded meetings with time-aligned transcriptions, speaker labels and dialogue act annotations. This gives you real conversations with real turn boundaries, which is what you need to learn when a turn ends. Published work on this corpus reports a turn-change ratio of roughly 64 percent shift to 36 percent hold, so you have a benchmark.
Silero VAD	Free, MIT licensed. A pre-trained voice activity detector. Use it to know when there is speech at all; your own model decides when the turn is over.
Your own recordings	Record at least 30 minutes of people talking to your system, including pauses, hesitations, and interruptions. This is your real test set.

Free tools for the parts you do not build

- Speech to text: faster-whisper running locally, or the browser speech API. Both free.
- The reply: any free-tier language model API. Keep the prompt short; long replies destroy the feeling of a conversation.
- Text to speech: Piper, which is free, fast and runs locally, or the browser speech synthesis.
- Hosting: Hugging Face Spaces or a free-tier host. Your choice of web or Android.

Day two checkpoint. Audio captured from a microphone in the browser or app, Silero VAD reporting speech and silence on it, and AMI downloaded. Post confirmation in the group.

3. The Work

Stage A - Build the naive version first, and feel what is wrong with it

- Wire the simplest thing that works: record, wait for 1 second of silence, transcribe, send to the model, speak the reply.
- Use it for ten minutes and write down what is wrong. It will cut you off mid-sentence and it will leave long gaps. Record how often each happens.
- Measure the delay from when you stop speaking to when it starts speaking. That number is what the rest of the project reduces.

Stage B - Build the endpointing model

- This is the centre of the project. Train a model that decides, continuously, whether the speaker has finished.
- Use AMI to build training examples: points inside a turn, and points at the end of a turn. Report how many of each you have.
- Feed it more than silence duration. Use the pitch and energy falling at the end of a phrase, how long the pause has run, and whether the words so far form a complete thought.
- Report two numbers that trade off against each other: how often it cuts the speaker off, and how long it waits when they have finished. Plot one against the other and choose an operating point.
- Compare against the fixed silence threshold from Stage A. If your model does not beat it, say so and show why.

Stage C - Handle interruption

- While the assistant is speaking, keep listening. When the user starts talking, stop the assistant immediately.
- The hard part is that the microphone hears the assistant’s own voice. Handle that, and say how.
- Measure how long it takes to stop after the user starts speaking. Anything over about half a second feels rude.

Stage D - Make it fast enough to feel live

- Measure every stage separately: speech to text, model reply, text to speech, and the endpointing decision. Put them in a table.
- Start speaking the reply before the whole reply is ready. Waiting for the full text before speaking is the largest single delay and the easiest to remove.
- State your target for total delay and whether you hit it. Under one second feels like conversation; over two seconds does not.

Stage E - Captions and the finished product

- Show captions as the assistant speaks, and show the user’s own words as they are recognised.

- Handle the ugly cases: no speech at all, background noise, the user saying nothing for a long time, the model failing. None of these may crash the conversation.
- Have at least ten people hold a real conversation with it. Report how many turns before something went wrong, and what went wrong.

5. What You Hand In

Five things. All five are required; a missing one is an incomplete submission.

Deliverable	Requirement
1. Working demo	A live URL anybody can open and use, or an installable APK. It must work on a phone. If it needs internet, say so; if it works offline, prove it.
2. Repository	Public GitHub repo under the Skye8 organisation. README that lets a stranger run it. No data files or credentials in git history.
3. Technical report	Four pages maximum. The problem, the data, what you built, how well it works, and what it gets wrong. Numbers, not adjectives.
4. Pitch	Five slides, five minutes, presented live. What it does, who buys it, why it is better than what they do now.
5. Business case	One page. Who pays, how much, what it costs to run per month, and how many customers before it breaks even. Use real figures you can source.

The business case is not decoration. If you cannot name a buyer in Bamenda and a price they would pay, you have built a demonstration, not a product.

6. Acceptance

1. A person can hold a spoken conversation with the demo, with captions, on a phone.
2. The endpointing model is trained by you, not replaced by a fixed silence timer.
3. Cut-off rate and waiting time are both reported, plotted against each other, with a chosen operating point.
4. Your model is compared against the fixed threshold baseline, honestly.
5. Interruption works and the time to stop speaking is measured.
6. Delay is broken down by stage in a table, with a stated target.
7. At least ten people used it and the failures are reported.

Everyone building this reaches for a silence timer, and everyone who does ends up with something that talks over people. The whole value of this project is in refusing that shortcut and measuring what you gained by refusing it.

7. Submission

Everything is submitted on the Skye8 platform. Nothing sent by email, WhatsApp or Drive link will be marked.

1. Submit the live demo URL or APK, the repository URL, and the commit hash of your final merge.
2. Attach the technical report, the slide deck and the business case.
3. Submit before 23:59 on Friday 2 October 2026. The platform timestamp is the only record.
4. Pitches are presented the following Monday evening in the online session.